



Name: [REDACTED]
 MRN: [REDACTED]
 D.O.B. (Age:)
 Sex: F
 Location:

Path No.: [REDACTED]
 Date Obtained:
 Date Received:
 Physician:

SURGICAL PATHOLOGY

****See Addendum/Procedure****

SPECIMEN:

- A: Subnipple tissue, right breast, excision
- B: Lymph node, right axilla sentinel node, biopsy
- C: Lymph node, right axilla sentinel node #2, biopsy
- D: Breast, right, total mastectomy
- E: Skin, right breast, excision

*ICD O.3
 Carcinoma, infiltrating duct NOS 8560/3
 Date of Breast NOS 650.9
 (R) Breast, central portion 650.1*

DIAGNOSIS(ES):

- A. Subnipple tissue, right breast, excision:
 Nipple ducts with atypical micropapillary hyperplasia. No carcinoma seen.
- B. Lymph node, right axilla sentinel node, biopsy:
 Lymph node, bisected and serially sectioned, negative for carcinoma. IHC for cytokeratin is negative.
- C. Lymph node, right axilla sentinel node #2, biopsy:
 Lymph node, quadrasected and serially sectioned, single focus of carcinoma seen by light microscopy and IHC for cytokeratin (focus of < 10 cells, ITC).
- D. Breast, right, total mastectomy:
 Infiltrating ductal carcinoma, well differentiated (1+2+1=4/9), spanning 2.0 cm, arising in a background of ductal carcinoma in situ, intermediate to high grade.
 Infiltrating ductal carcinoma, separate focus, moderately differentiated (3+3+1=7/9), spanning 0.6 cm.
 Extensive ductal carcinoma in situ, solid, cribriform and micropapillary type, with necrosis and microcalcification, seen on 15 of 26 slides and extending to within 0.1 cm of deep margin. The micropapillary DCIS is seen in the nipple duct sections.
 Lobular carcinoma in situ, monomorphic type.
 Intraductal papillomas and sclerosing adenosis with microcalcification.
 Fibroadenoma
- E. Skin, right breast, excision:
 Fragments of skin.

CLINICAL INFORMATION: Right breast cancer.

GROSS DESCRIPTION:

The specimen is received in five parts.

Part A is received fresh from the operating room for frozen section diagnosis in a container labeled with the patient's name and "Rt subnipple tissue R/O cancer". It consists of one piece of red soft tissue measuring 1.1 x 0.9 x 0.3 cm.

The specimen is entirely submitted in one cassette labeled AFS.

Part B is received fresh from the operating room for frozen section diagnosis in a container labeled with the patient's name and "Right axillary sentinel lymph node #1". It consists of one piece of fatty tissue measuring 3.2 x 1.4 x 0.8 cm. One lymph node is palpated measuring 1.8 cm in greatest dimension. The lymph node is bisected. The specimen is entirely submitted in four cassettes labeled BFS1-BFS2, B1 and B2. Please note: Three levels and cytokeratin stains are ordered for the lymph node only.

Legend: BFS1-BFS2 = frozen section-lymph node, B1-B2 = remaining tissue.

Part C is received fresh from the operating room for frozen section diagnosis in a container labeled with the patient's name and "Right axillary sentinel lymph node #2". It consists of one piece of fatty tissue measuring 2.8 x 1.9 x 1.1 cm. One lymph node is palpated measuring 2.7 cm in greatest dimension. The lymph node is quadrisectioned. The specimen is entirely submitted in five cassettes labeled CFS1-CFS4 and C1. Please note: Three levels and cytokeratin stains are ordered for the lymph node only.

Legend: CFS1-CFS4 = frozen section-lymph node, C1 = remaining tissue.

Part D is received unfixed in a container labeled with the patient's name and "Right breast mastectomy, stitch short anterior subnipple tissue, stitch long = lateral". It consists of a right total skin sparing mastectomy specimen measuring 22.0 x 14.0 x 3.5 cm. A short suture is noted indicating the subareolar and a long suture is noted indicating the superior (as per Dr. , the long suture was mistakenly labeled "lateral"). An underlying mass is palpable located 1.0 cm from the subareolar. The subareolar is inked blue, the superficial margin is inked yellow and the deep margin is inked black. The axillary tissue measures 2.5 x 1.5 x 0.5 cm. No lymph nodes are palpable within it.

The specimen is serially sectioned at closely spaced intervals, revealing an area of white, cystic parenchyma measuring 6.0 x 4.0 x 2.5 cm located in the center, 0.5 cm from the deep margin and 0.2 cm from the superficial margin. Within the cystic parenchyma is a firm, poorly circumscribed, white-tan mass measuring 2.0 x 2.0 x 1.5 cm which comes to within 0.6 cm from the deep margin and 1.3 cm from the superficial margin. At least four scattered yellow-white nodules measuring from 0.5 cm to 0.8 cm in diameter are also seen within the cystic parenchyma located 0.3 cm from the deep margin and 1.5 cm from the superficial margin and at least 1.0 cm from the mass. The remainder of the specimen is composed of a moderate amount of breast tissue (60%) and a moderate amount of yellow fatty tissue (40%). Also received in the container is one irregular piece of soft, fibrofatty, yellow-red tissue measuring 4.0 x 3.0 x 1.0 cm. On cut surface, no discrete lesions or lymph nodes are identified. Representative sections are submitted in 26 cassettes labeled D1-D26. Please note: The specimen was placed in formalin at

LEGEND:

D1-D2 = Submucosal area

D3-D4 = Mass with adjacent deep margin

D5-D7 = Remaining portion of mass

D8 = Superficial margin closest to mass

D9-D10 = Nodule closest to mass with adjacent surrounding tissue and deep margin

D11 = Remaining portion of nodule with adjacent surrounding tissue

D12-D15 = Surrounding nodules in cystic parenchyma

D16-D17 = Cystic parenchyma

D18-D19 = Upper inner quadrant

D20-D21 = Lower inner quadrant

D22-D23 = Lower outer quadrant

D24-D26 = Upper outer quadrant

D26 = Fibrofatty tissue also received in the container.

Part E is received unfixed in a container labeled with the patient's name and "Skin from mastectomy R breast". It consists of a strip of rubbery, tan-brown skin measuring 11.6 x 0.7 x 0.2 cm. Representative sections are submitted in one cassette labeled E1. Please note: The specimen was placed in formalin at

INTRAOPERATIVE CONSULTATION:

AFS: Benign breast tissue and large ducts. No carcinoma seen.

BFS1, BFS2: No carcinoma seen.
CFS1, CFS2: No carcinoma seen.
CFS3, CFS4: No carcinoma seen.

Performed by: Resident:
Interpreted by: Attending:

MICROSCOPIC DESCRIPTION:

- I. TYPE OF SPECIMEN: Right total mastectomy with sentinel node dissection.
- II. LOCATION OF THE TUMOR: Central.
- III. TYPE OF NEOPLASM: Carcinoma, Invasive, Ductal - NOS (Nodule 1)
Well Differentiated, Total score 4
(Tubule Score 1, Nuclear Grade Score 2, Mitotic Score 1).

TUMOR SIZE: The tumor's greatest dimension 2.0 cm.

Carcinoma, Invasive, Ductal - NOS (Nodule 2)
Moderately Differentiated, Total score 7.
(Tubule Score 3, Nuclear Grade Score 3, Mitotic Score 1).

TUMOR SIZE: The tumor's greatest dimension 0.6 cm.

Ductal carcinoma in situ, nuclear grade 3, widespread 30%.

QUANTITATION:

Ductal carcinoma in situ is present in 15 of 26 slides of breast.

Intraductal solid subtype, cribriform, comedo, micropapillary/

- VI. VASCULAR SPACE INVASION: Present in lymphatics.
- VII. CALCIFICATION: Present in both malignant and benign areas.
- VIII. NIPPLE: Ducts involved by DCIS.
- IX. SKIN: Present, uninvolved by cancer.
- X. ADJACENT BREAST TISSUE: Cystic disease, proliferative with atypia.
- XI. MARGINS: Tumor distance from closest margin DCIS- <0.1 cm from deep margin.
- XII. AXILLARY LYMPH NODES: Sentinel - 2.
- XIII. POSITIVE LYMPH NODES: Sentinel - ITC only.
- XV. PATHOLOGIC STAGING (pTNM) AJCC 7th Edition: *Reflects staging only of the current specimen. Ultimate staging responsibility rests with the primary physician.*

pT1c: Tumor more than 1.0 cm but not more than 2.0 cm in greatest dimension.

pN0(ITC,+): No regional lymph node metastasis on H & E histologically, (ie. Individual tumor cells seen, none greater than 0.2 mm, positive IHC).

COMMENT(S):

Case shown at

This report has been reviewed electronically and signed on by

Interpreted by: Attending:

The diagnosis was rendered by the attending pathologist.

Receptor Expression Analysis

Date Ordered:

Status: Signed Out

INTERPRETATION

Test Performed on formalin fixed paraffin embedded section of: mastectomy

The results are for invasive carcinoma #1

Specimen part: D

Slide#: D3

Fixation: adequate.

Results:

This tumor is considered positive for estrogen receptor expression (≥ 1 % of the cells are positive). The specimen was adequate for evaluation for ER as per ASCO/CAP guidelines 2010. **Approximately 75 % of the carcinoma cell nuclei stain with an immunohistochemical stain utilizing an anti-estrogen receptor antibody () with an average intermediate intensity.**

This tumor is considered positive for progesterone receptor expression (≥ 1 % of the cells are positive). The specimen was adequate for evaluation for PR as per ASCO/CAP guidelines 2010. **Approximately 80 % of the carcinoma cell nuclei stain with an immunohistochemical stain utilizing an anti-progesterone receptor antibody () with an average strong intensity.**

Equivocal/indeterminate for HER2 expression. Her-2/neu over-expression has been evaluated, on a formalin fixed paraffin embedded section, using the (proprietary kit). Using the ASCO/CAP scoring criteria, the score is: 2.

Invasive carcinoma #2:

This tumor is considered positive for estrogen receptor expression (≥ 1 % of the cells are positive). The specimen was adequate for evaluation for ER as per ASCO/CAP guidelines 2010. **Approximately 100 % of the carcinoma cell nuclei stain with an immunohistochemical stain utilizing an anti-estrogen receptor antibody SP1) with an average strong intensity.**

This tumor is considered positive for progesterone receptor expression (≥ 1 % of the cells are positive). The specimen was adequate for evaluation for PR as per ASCO/CAP guidelines 2010. **Approximately 100 % of the carcinoma cell nuclei stain with an immunohistochemical stain utilizing an anti-progesterone receptor antibody 1E2) with an average strong intensity.**

Positive for HER2 overexpression. Her-2/neu over-expression has been evaluated, on formalin fixed paraffin embedded sections, using the (proprietary kit). Using the ASCO/CAP scoring criteria, the score: 3.

This procedure/addenda has been electronically reviewed and signed on

by.

Interpreted by: Attending:

Criteria	Yes	No
Diagnosis Discrepancy		/
Primary Tumor Site Discrepancy		/
HIPAA Discrepancy		/
Prior Malignancy History		/
Dual/Synchronous Primary Noted		/
Case is (circle):	QUALIFIED	DISQUALIFIED
Reviewer Initials	W	1/20/14

Note: Immunohistochemistry testing performed at was developed and its performance characteristics determined by the

These tests were interpreted in conjunction with external positive and internal negative controls, unless otherwise noted, it has not been cleared or approved by the US FDA. This test is used for clinical purposes only. It should not be regarded as investigational or for research.

END OF REPORT